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1. What is the primary difference between classic machine learning and deep network-based machine learning in computer vision?

The primary difference is way they represent and process visual data. Classic machine learning approaches rely on hand-engineered features, such as edges, corners, and shapes, to describe images. In contrast, deep network-based machine learning uses multiple layers of artificial neural networks to automatically learn and represent complex features from raw image data. This allows deep networks to learn hierarchical representations of images, from simple edges to complex objects, and achieve concurrent performance in various computer vision tasks.

2. What are the factors that enabled the deep network revolution?

First, the availability of large-scale datasets, such as ImageNet, provided the necessary data to train deep networks. Second, advances in computing hardware, particularly the development of graphics processing units (GPUs), enabled fast and efficient computation of complex neural networks. Third, the introduction of new techniques, such as convolutional neural networks (CNNs), rectified linear units (ReLUs), and batch normalization, improved the training and performance of deep networks. Finally, the development of open-source software frameworks, such as TensorFlow and PyTorch, made it easier for researchers and practitioners to implement and experiment with deep networks.

3. Consider the architectures of GoogleNet, VGG-16, and ResNet52. Which network has more layers that can integrate information from the entire image?

ResNet52 has more layers, and it uses a technique called residual connections, which allows the network to learn features from the entire image by adding the output of previous layers to the current layer. This enables the network to capture long-range dependencies and integrate information from different parts of the image. In contrast, GoogleNet and VGG-16 use convolutional and pooling layers to process the image, but they do not have the same level of integration across layers.

4. What is a residual connection and how does it change the function a particular layer of the network is learning?

A residual connection is a technique used in deep neural networks where the output of a previous layer is added to the output of the current layer. This changes the function a particular layer is learning by allowing it to learn the residual, or the difference, between the input and the output of the previous layer.

5. In Files::CV Papers, find the paper [huang-DenseNet-CVPR17.pdf](#), which describes a network called DenseNet. Focus on the architecture diagram. DenseNet uses skip connections rather than residual connections. How does the DenseNet architecture combine filters from different layers of the block?

DenseNet concatenates the feature maps of all previous layers to the current layer. This allows the network to capture a wide range of features and reuse them across different layers.

6. Within a DenseNet block, does DenseNet reduce the number of filters that need to be computed compared to an equivalent number of layers for a ResNet architecture? How does it do that?

Yes, DenseNet achieves this by reusing features from previous layers through skip connections, which reduces the number of parameters and computations required. Additionally, DenseNet uses a technique called growth rate, which controls the number of new filters added at each layer, allowing the network to grow gradually and efficiently.

7. What is the difference between a residual connection and a skip connection?

A residual connection adds the output of a previous layer to the output of the current layer, whereas a skip connection concatenates the feature maps of all previous layers to the current layer. Residual connections are used in ResNet architectures, while skip connections are used in DenseNet architectures. Both techniques aim to alleviate the vanishing gradient problem and enable the network to learn more complex features, but they differ in how they combine information from previous layers.

8. Why do residual and skip connections facilitate much deeper networks?

Residual and skip connections facilitate much deeper networks by alleviating the vanishing gradient problem, which occurs when gradients become smaller as they are backpropagated through the network. By adding or concatenating the output of previous

layers to the current layer, these connections allow the network to learn more complex features and capture long-range dependencies. This enables the network to grow deeper without suffering from the vanishing gradient problem, leading to improved performance on various tasks.